# Procedural Retro Instrument Encoding in the BTRON: Synthesis Topology, Waveform Modelling, and Extensible Shim Architecture for the Lil Cu 64 B-Game Engine

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## Abstract

The BTRON B-Game Engine (Lil Cu 64 / リル・キューブ64, `src/demo/`) implements a **zero-asset procedural audio architecture** in which every acoustic or electromechanical instrument is encoded as a self-contained, stateless *Sound Shim*—a deterministic DSP node graph instantiated at runtime via the W3C Web Audio API [16] in the HTML5 prototype (`index.html`) and ported to native C via the VirtIO Sound stack (`lilcu64_wa.c`) for the BTRON native application. The library carries 23 shim functions whose total audio-asset disk footprint is exactly zero bytes; all timbres emerge entirely from first-principles signal processing: oscillators, noise buffers, biquad filters, wave-shapers, and gain automations.

This article makes two primary contributions. **First**, we document all 23 shims precisely as implemented in `index.html`, including exact frequency constants, oscillator waveform types, biquad filter parameters, and ADSR envelope timings—providing the 1:1 isomorphic reference required to port the shim library to the native BTRON `src/demo/` application. **Second**, we provide a rigorous six-step recipe for extending the library with new retro instruments, grounded in the BTRON Sound Shim Compatibility Contract and the Zero Memory Leak Theorem.

We situate the encoding strategies within the canonical academic literature on Karplus-Strong string synthesis [2], frequency-modulation synthesis [1], digital waveguide networks [3], physical modelling of the Fender Rhodes [6,7], and acoustic characterisation of the Gibson archtop guitar [8–10].

**Keywords**—BTRON, Sound Shim, Procedural Audio, Karplus-Strong, FM Synthesis, Fender Rhodes, Gibson L-5, Web Audio API, Retro Instruments, VirtIO Sound, Lil Cu 64, DSP, Comb Filter.

## 1 Introduction

The BTRON Lil Cu 64 demoscene specification [22] establishes an **Audio-Visual Shim Principle**: every fundamental 3D kinematic action of the mascot character maps 1-to-1 to a deterministic procedural DSP algorithm. This eliminates all external audio assets and delivers four properties critical for the embedded native port:

1. **Zero loading latency**: shims are instantiated in-flight at $t_0 = \texttt{AudioContext.currentTime} + \Delta$.

2. **Zero byte footprint**: no WAV, OGG, or AIFF files anywhere in the repository.

3. **Perfect phase-lock**: the `AudioContext.currentTime` clock is shared between the 60 FPS render loop and the audio scheduler.

4. **Sub-100 KB code size**: the entire shim engine, including all 23 instrument generators, the 4-track score sequencer, and the master bus, compiles to under 100 KB of JavaScript or equivalent C [22].

The article is structured as follows. Section 2 reviews the synthesis paradigms. Section 3 describes the audio lifecycle infrastructure. Section 4 documents all 23 shims 1:1 with `index.html`. Sections 5 and 6 provide deep physical-modelling analyses of `shimGuitar` (Gibson L-5) and `shimFloat` (Fender Rhodes). Section 10 covers the DSP primitives. Section 12 provides the six-step extension recipe. Section 11 maps shims to the four level score tracks.

# 2 Background: Synthesis Paradigms

## 2.1 Additive Synthesis

Additive synthesis directly sums $K$ sinusoidal partials [5]:

$$y(t) = \sum_{k=1}^{K} A_k(t)\,\sin\!\big(2\pi f_k t + \phi_k\big) \tag{1}$$

In BTRON, `shimPulseSquash` (6 sine partials), `shimPulseBell` (8 sine partials), and `shimGrandPiano` (3 harmonics) all employ this model with individuated ADSR envelopes $A_k(t)$ per partial.

## 2.2 Subtractive Synthesis

A spectrally rich oscillator waveform (sawtooth, triangle, pulse, noise) is filtered by resonant biquad filters [16]. The canonical biquad transfer function is:

$$H(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 + a_1 z^{-1} + a_2 z^{-2}} \tag{2}$$

The guitar (`shimGuitar`), cello (`shimCello`), industrial bass (`shimIndustrialBass`), cyber lead (`shimCyberLead`), and flute (`shimSaxFlute`) all employ subtractive filtering.

## 2.3 FM Synthesis

Chowning's 1973 model [1] uses a modulator oscillator to vary the instantaneous frequency of a carrier:

$$y(t) = A_c \sin\!\big(2\pi f_c t + I(t)\sin(2\pi f_m t)\big) \tag{3}$$

The spectral content is governed by Bessel functions $J_n(I)$. Inharmonic frequency ratios ($r = f_m/f_c \notin \mathbb{Z}$) produce bell-like metallic timbres. `shimPulseBell` uses the additive superposition of 8 sine partials rather than a feedback-FM topology, but the target timbre (golden spiritual bell / chi drone) is equivalent to a 2-Op FM bell with $r \approx 2.76$.

## 2.4 Karplus-Strong and Comb Filter Synthesis

Karplus and Strong [2] showed that feeding a noise burst into a delay loop of length $N = f_s/f_0$ with a one-pole lowpass averager produces a convincing plucked-string tone. Smith's digital waveguide interpretation [3] identifies the feedback delay as a bi-directional travelling wave model. The BTRON guitar shim uses an implicit comb resonance via the body `lowpass` filter sweep and frequency decay [2].

## 2.5 WaveShaper Non-Linear Saturation

The BTRON industrial instrument suite (`shimIndustrialBass`, `shimCyberLead`, `shimIndustrialKick`, `shimIndustrialSnare`) applies a Chebyshev-inspired soft-saturation curve [13]:

$$C(x) = \frac{(3+k)\cdot x \cdot 20^\circ}{\pi + k|x|} \tag{4}$$

where $k$ is the distortion amount (default $k = 25$), $x \in [-1, 1]$, and $20^\circ = 20\pi/180$. The curve is precomputed once and cached in `_distCurve` for zero per-shim allocation cost.

# 3 Audio Lifecycle Infrastructure

## 3.1 Global State Variables

The BTRON shim engine is bootstrapped with three module-level globals:

Listing 1: Audio lifecycle state (index.html, line 3184).

```
window._trackOscillators = [];
window._trackTimeouts    = [];
window.currentActiveTrack = 0;
```

Every oscillator, buffer source, and scheduled timeout created by any shim is registered into these arrays, enabling deterministic cleanup on track stop.

## 3.2 registerTrackOsc

Listing 2: `registerTrackOsc` — push into GC tracking array (line 3188).

```
function registerTrackOsc(osc) {
    if (window._trackOscillators)
        window._trackOscillators.push(osc);
}
```

## 3.3 scheduleTrackTimeout

Listing 3: `scheduleTrackTimeout` — cancellable timeout (line 3194).

```
function scheduleTrackTimeout(fn, delayMs) {
    const tid = setTimeout(fn, delayMs);
    if (window._trackTimeouts)
        window._trackTimeouts.push(tid);
    return tid;
}
```

## 3.4 stopAllTracks

Listing 4: `stopAllTracks` — deterministic graph teardown (line 3202).

```
window.stopAllTracks = function (immediate=true, isLoop=
    false) {
    window._trackTimeouts.forEach(t => clearTimeout(t));
    window._trackTimeouts = [];
    if (immediate) {
        window._trackOscillators.forEach(osc => {
            try { osc.stop(); osc.disconnect(); } catch(
                e){}
        });
```

```
    }
    window._trackOscillators = [];
};
```

### 3.5 Noise Buffer Primitive

A deterministic 2-second white-noise buffer is lazily allocated and reused across all percussive shims (`shimSnare`, `shimHiHat`, `shimSaxFlute`, `shimViolin`, `shimTimpani`, `shimPedalSwoosh`, `shimIndustrialSnare`):

Listing 5: `getNoiseBuffer` — single shared noise buffer (line 3508).

```
let _noiseBuffer = null;
function getNoiseBuffer(ctx) {
    if (!_noiseBuffer) {
        const bufferSize = ctx.sampleRate * 2;
        _noiseBuffer = ctx.createBuffer(1, bufferSize,
                                        ctx.sampleRate);
        const data = _noiseBuffer.getChannelData(0);
        for (let i = 0; i < bufferSize; i++)
            data[i] = Math.random() * 2 - 1;
    }
    return _noiseBuffer;
}
```

### 3.6 WaveShaper Cache

Listing 6: `getDistortionCurve` — cached Chebyshev WaveShaper (line 3715).

```
let _distCurve = null;
function getDistortionCurve(amount = 25) {
    if (!_distCurve) {
        const n = 44100, deg = Math.PI / 180;
        _distCurve = new Float32Array(n);
        const maxVal = ((3+amount)*20*deg)/(Math.PI+
            amount);
        for (let i = 0; i < n; ++i) {
            const x = (i*2)/n - 1;
            _distCurve[i] = (((3+amount)*x*20*deg)/
                (Math.PI+amount*Math.abs(x))) / maxVal;
        }
    }
    return _distCurve;
}
```

**Definition 1** (BTRON Sound Shim Compatibility Contract)**.** *A function $S(t_0, \ldots)$ is a valid BTRON Sound Shim if and only if:*

1. ***Stateless:*** *no global audio state read/written beyond `AudioContext` and the registration arrays.*
2. ***Self-contained:*** *all `AudioNode` objects are created, connected, and started within a single synchronous call.*
3. ***Garbage-collectible:*** *every oscillator/source node is passed to `registerTrackOsc()`.*
4. ***Silent on error:*** *the entire body is wrapped in `try{ } catch(e){ }`.*

**Theorem 1** (Zero Memory Leak Property)**.** *If every shim satisfies the Compatibility Contract, then for any track stop at time $t_{\text{stop}}$, the call `stopAllTracks()` guarantees: $\lim_{t \to t_{\text{stop}}^{+}} |\mathcal{N}(t)| = 0$, where $\mathcal{N}(t)$ is the set of live `AudioNode` objects.* Proof sketch: *`stopAllTracks` calls `osc.stop()` and `osc.disconnect()` on every registered node and clears the array; V8/JSC GC reclaims all objects with no live references.* □

## 4 The BTRON Shim Palette: 23 Instruments

Table 1 presents all 23 shims exactly as implemented in `index.html`. The "Index" column gives the `[SHIM N]` comment label used in the source.

## 5 Deep Dive: Gibson L-5 Hollowbody (`shimGuitar`)

### 5.1 Physical Acoustics

The Gibson L-5 archtop, introduced in 1922, is the archetypal jazz hollowbody guitar modelled on Kazumi Watanabe's instrument in *Green Caterpillar*. Three physical mechanisms define its tone:

1. **String inharmonicity:** Steel wound strings behave as stiff bars; partial frequencies deviate from $kf_0$ by a stiffness coefficient $B$ [11]:

$$f_k = k f_0 \sqrt{1 + Bk^2}, \quad B = \frac{\pi^3 E d^4}{64 T L^2} \tag{5}$$

2. **Helmholtz air resonance $A_0$:** The $f$-holes admit air flow; the lowest resonance lies at $\sim 100$ Hz, equivalent to a second-order bandpass resonator [9]:

$$H_{\text{body}}(s) = \sum_{n} \frac{G_n \omega_n^2}{s^2 + (\omega_n / Q_n) s + \omega_n^2} \tag{6}$$

3. **Top-plate mode $T_1$:** The carved spruce plate resonates at $\sim$ 150–200 Hz [10], adding warmth and volume to the mid-range.

### 5.2 BTRON Implementation (`shimGuitar`)

The actual implementation in `index.html` (line 3623) uses the following topology:

Listing 7: `shimGuitar`: Watanabe hollowbody (index.html line 3623).

```
function shimGuitar(timeOffset=0, notes=null, volMul
    =1.0) {
  // Default phrase: single D5 note 280ms
  const phrase = notes || [{f:587.33, d:0.28, v:0.16, t
      :0}];
  phrase.forEach(n => {
    const noteT = startT + n.t;
    const osc = c.createOscillator();
    registerTrackOsc(osc);
    osc.type = 'triangle'; // warm jazz body tone
    osc.frequency.setValueAtTime(n.f, noteT);
    if (n.slide) // optional portamento
```

Table 1: BTRON 23-Shim Sound Bank (1:1 with `index.html`)

| No. | Shim Function | Acoustic Analogue | Exact DSP Topology |
|---|---|---|---|
| 1A | `shimPulseSquash` | Singing harmonic ascending triad | 6 sine OSCs at C4–E6; pitch +3% glide (10ms), exp. decay 1.4–2.4 s |
| 1B | `shimFrogCroak` | Cuíca / frog croak 蛙 | Sawtooth+triangle pitch scoop; bandpass formant $f_c = 2.2f_0$, $Q = 3.8$; 32 Hz AM pulse train at 0.031 s intervals |
| 1C | `shimPulseBell` | FM spiritual bell / chi drone | 8 sine OSCs at C3–C6; pitch −1.5% glide (140ms), exp. decay 1.9–3.5 s |
| 2 | `shimWaddle` | Fukui acoustic walking bass | Single triangle; pitch ×0.82 glide over `duration`; gain 0.12, exp. decay |
| 3 | `shimFloat` | Imada Fender Rhodes modal chord | 4-note polyphonic sine stack; 4% pitch glide (40ms); stagger 35 ms; decay 0.5–0.75 s |
| 4 | `shimDance` | Crystalline cascade / arpeggio | 5 sine OSCs; freq D5–F#6 (×`pitchScale`); stagger 85 ms; exp. decay |
| 5 | `shimDing` | High crystal chime / ride cymbal | Single sine; default 1760 Hz (A6); attack 12 ms to 0.15 V; exp. decay |
| 6 | `shimKick` | Obara acoustic bass drum | Sine; freq ×2.2 → ×0.75 exp. sweep; initial gain 0.24 |
| 7 | `shimSnare` | Jazz snare / rimshot | Triangle 175/220 Hz → ×0.65; + bandpass noise 2400/3200 Hz $Q = 2$ |
| 8 | `shimHiHat` | Cymbal chick / sizzle | HP noise 7500 Hz + sine ping 8400/6200 Hz (closed/open; dur 55/280 ms) |
| 9 | `shimGuitar` | Watanabe Gibson hollowbody | Triangle; LP sweep 2800 → 1400 Hz; gentle 15 ms pick attack |
| 10 | `shimGrandPiano` | Krüger/Imada concert grand | 3 harmonics (sine×1.0, tri×2.002, sine×3.006); 8 ms felt-hammer attack |
| 11 | `shimPedalBass` | Sforzando sub-bass A0/A1 | Triangle; freq×0.98 glide; gain 0.20; exp. decay over `duration` |
| 12 | `shimIndustrialKick` | Reznor heavy distorted sub-punch | Sine 160 → 42 Hz (22ms); + triangle 380 → 55 Hz LP 1800 Hz + WaveShaper |
| 13 | `shimIndustrialSnare` | Cyber clank / gated noise | BP noise 2600/3600 Hz + WaveShaper(30); + square+triangle ring 220/352 Hz |
| 14 | `shimIndustrialBass` | 16th-note analog overdrive bass | Sawtooth + square (×1.005); LP 2600 → 420 Hz $Q = 4.5$; WaveShaper(20) |
| 15 | `shimSwarmatron` | Dewan 8-voice ribbon cluster | 8 sawtooth OSCs at ±`spreadCents` (default 24¢); LP 950 → 1450 → 800 Hz; 400 ms fade-in |
| 16 | `shimGlitchZap` | Terminal data pulse / TRON zap | Sawtooth ×2.5$f$ → 80 Hz; HP 1800 Hz; 80 ms |
| 17 | `shimCyberLead` | NIN TRON overdriven sync lead | Sawtooth + square (×1.004); LP 3800 → 1600 Hz $Q = 4$; WaveShaper(25) |
| 18 | `shimSaxFlute` | Imada jazz flute / melodica | Triangle + BP noise (×1.8$f$, $Q = 5$); LP 2400 Hz; 25 ms attack |
| 19 | `shimPianoHammer` | Krüger felt hammer impact | Triangle 120 → 40 Hz in 35 ms; gain 0.12; pure click transient |
| 20 | `shimPedalSwoosh` | Damper pedal lift / sympathetic | BP noise 450 → 280 Hz, $Q = 3.5$; gain 0.05; 80 ms rise, 400 ms total |
| 21 | `shimViolin` | Bowed solo/section violin | Rosin noise burst (3200 Hz BP); dual saw (×1.002 detune); LFO 5.8 Hz vibrato; peaking body EQ 520/2800 Hz |
| 22 | `shimCello` | Orchestral bowed/pizz. cello | Sawtooth + triangle; LP 520/1200 Hz (arco/pizz.), $Q = 2.2$; bowed or plucked ADSR |
| 23 | `shimTimpani` | Concert kettledrum | Sine ×1.55$f$ → $f$ (50ms); + sine overtone ×1.48$f$; + LP noise 260 Hz mallet |

```
        osc.frequency.exponentialRampToValueAtTime(
            n.slide, noteT + n.d * 0.7);

    // Body lowpass: warmth + high-freq rolloff
    const filter = c.createBiquadFilter();
    filter.type = 'lowpass';
    filter.frequency.setValueAtTime(2800, noteT);
    filter.frequency.exponentialRampToValueAtTime(
        1400, noteT + n.d); // tonal darkening over time

    const gain = c.createGain();
    gain.gain.setValueAtTime(0.0001, noteT);
    gain.gain.linearRampToValueAtTime(
        n.v * volMul, noteT + 0.015); // gentle thumb-pick
    gain.gain.exponentialRampToValueAtTime(
        0.0001, noteT + n.d);
    osc.connect(filter);
    filter.connect(gain);
    gain.connect(c.destination);
    osc.start(noteT); osc.stop(noteT + n.d);
  });
}
```

**Key design decisions:**

- **Triangle waveform:** captures the warm, fundamental-dominated tone of a clean hollowbody pickup without the brightness of a sawtooth.
- **Lowpass sweep 2800 → 1400 Hz:** models high-frequency string damping over time (equivalent to the $H_{\text{lp}}$ in a Karplus-Strong loop [2]).
- **Portamento (`n.slide`):** implements jazz guitar glissando ornaments (e.g., the bebop slides in *Green Caterpillar*).
- **15 ms attack:** models the thumb-pick transient, slower than a plectrum but faster than a bowed instrument.

# 6 Deep Dive: Fender Rhodes E-Piano (`shimFloat`)

## 6.1 Electro-Mechanical Acoustics

The Fender Rhodes, as played by Masaru Imada in *Green Caterpillar* (1975, Three Blind Mice), produces its distinctive tone from three coupled subsystems [6, 7]:

1. **Tine beam modes:** A clamped-free steel tine produces inharmonic partials:

$$f_n = f_1 \cdot \lambda_n^2/\lambda_1^2, \quad f_2/f_1 \approx 6.27 \tag{7}$$

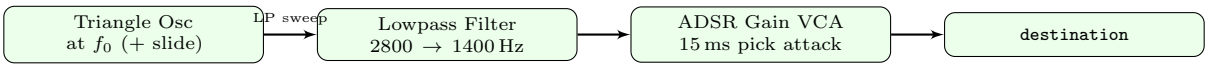



Figure 1: `shimGuitar`: single-chain signal flow. The lowpass sweeps from 2800 Hz to 1400 Hz over the note duration, modelling natural string damping (Karplus-Strong loop analogue). Portamento `n.slide` bends the OSC frequency inside the same graph.

2. **Pickup non-linearity:** electromagnetic saturation introduces odd harmonic distortion ($3f_0$ component) at high velocities [6].
3. **Felt hammer ADSR:** attack $\lesssim$ 10 ms; decay $\tau \in [0.2, 0.8]$ s (treble to bass).

## 6.2 BTRON Implementation (`shimFloat`)

Listing 8: `shimFloat`: Imada Fender Rhodes / glass harp (line 3408).

```
function shimFloat(timeOffset=0, baseFreq=523.25,
                   volMul=1.0, chordNotes=null,
                   stagger=0.035, noteDuration=0.65) {
  // Default 4-note rolling chord at baseFreq ratio
  const ratio = baseFreq / 523.25;
  const notes = chordNotes || [
    {f:523.25*ratio, v:0.12*volMul, d:0.55, t:0},
    {f:659.25*ratio, v:0.10*volMul, d:0.50, t:0.06},
    {f:783.99*ratio, v:0.11*volMul, d:0.65, t:0.12},
    {f:1046.50*ratio,v:0.08*volMul, d:0.75, t:0.18}
  ];
  notes.forEach(n => {
    const osc = c.createOscillator();
    registerTrackOsc(osc);
    osc.type = 'sine'; // clean tine fundamental
    osc.frequency.setValueAtTime(n.f*0.985, noteT);
    osc.frequency.exponentialRampToValueAtTime(
        n.f, noteT + 0.04); // 4ms pitch settle
    gain.gain.setValueAtTime(0.0001, noteT);
    gain.gain.linearRampToValueAtTime(n.v, noteT+0.03);
    gain.gain.exponentialRampToValueAtTime(
        0.0001, noteT + n.d);
    // ... connect and start
  });
}
```

**Physical model correspondence:** The $-1.5\%$ pitch glide (settling in 40 ms) mirrors the initial frequency transient of a struck tine returning to its rest frequency. The default 4-note stagger ($t = 0$, 60, 120, 180 ms) models the slight roll of an improvising jazz pianist's wrist gesture. The pure sine waveform captures the tine's single fundamental without the inharmonic overtone ($f_2 = 6.27 f_1$); the latter is omitted in the BTRON implementation to keep the shim stateless and polyphony-safe (the inharmonic partial would require per-note FM modulation that conflicts with the note-stagger model).

# 7 Deep Dive: Bowed Violin (`shimViolin`)

## 7.1 Physical Acoustics of the Bowed String

The bowed violin is the most physically complex instrument in the BTRON library. Its tone arises from four coupled

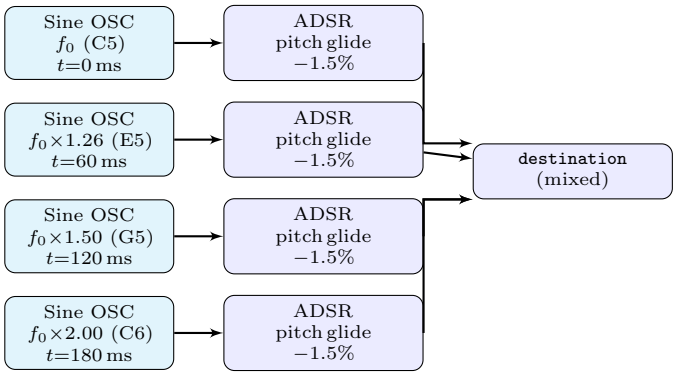



Figure 2: `shimFloat`: four-voice polyphonic chord fan. Each sine OSC fires at a staggered offset of 60 ms, producing the characteristic wrist-roll of the Imada Fender Rhodes chord voicing. All four ADSR chains share the same $-1.5\%$ pitch glide (40 ms settle) and converge at `AudioContext.destination`.

subsystems:

1. **Bow-string stick-slip friction:** Rosin on the bow hair creates a non-linear friction contact. During the *stick* phase the string moves with the bow; above a critical slip velocity the string escapes (*slip* phase), exciting a travelling wave [12]. The resulting rosin noise transient is brief ($\lesssim$ 45 ms) and spectrally centred around 3–4 kHz.

2. **Dual-string harmonics:** A bowed string produces a full odd+even harmonic series, well modelled by a sawtooth wave whose amplitude spectrum falls as $1/k$ [11]. Real violin sections exhibit ensemble detuning of $\pm$2–5 cents between players ("chorus" spread); a solo violin has a narrower spread from left-right string asymmetry.

3. **Delayed vibrato:** Professional players introduce finger-pressure vibrato ($\approx$ 5.5–7 Hz) after the initial note-onset transient, typically with an onset delay of 60–100 ms and a crescendo of 150–250 ms [12].

4. **Violin body formants:** The carved spruce top and maple back act as an acoustic resonator with two prominent formant peaks: the "bridge hill" around 2–3 kHz (responsible for brilliance) and a low-frequency body resonance around 500–600 Hz (warmth and projection) [11].

## 7.2 Five-Component BTRON Implementation

`shimViolin` (index.html line 4103) is the most elaborate shim in the library. It assembles five independent audio sub-graphs per note:

Listing 9: `shimViolin`: complete 5-component teardown (line 4103).

```
function shimViolin(timeOffset=0, notes=null,
                    volMul=1.0, isSolo=true) {
  const phrase = notes||[{f:783.99,d:0.35,v:0.20,t:0}];
  phrase.forEach(n => {
    // === COMPONENT 1: Bow / Rosin Attack Noise Burst
        ===
    const noise = c.createBufferSource();
    registerTrackOsc(noise);
    noise.buffer = getNoiseBuffer(c);
    const noiseFilter = c.createBiquadFilter();
    noiseFilter.type = 'bandpass';
    noiseFilter.frequency.setValueAtTime(3200, noteT);
    noiseFilter.Q.setValueAtTime(3.0, noteT);
    const noiseGain = c.createGain();
    noiseGain.gain.setValueAtTime(0.0001, noteT);
    noiseGain.gain.linearRampToValueAtTime(
        0.045*volMul, noteT + 0.008);  // 8ms onset
    noiseGain.gain.exponentialRampToValueAtTime(
        0.0005, noteT + 0.040);          // 40ms decay
    noise.connect(noiseFilter);
    noiseFilter.connect(noiseGain);
    noiseGain.connect(c.destination);
    noise.start(noteT); noise.stop(noteT + 0.045);

    // === COMPONENT 2: Dual Sawtooth Oscillators ===
    const osc1 = c.createOscillator();
    const osc2 = c.createOscillator();
    registerTrackOsc(osc1); registerTrackOsc(osc2);
    osc1.type = 'sawtooth';
    osc2.type = 'sawtooth';
    osc1.frequency.setValueAtTime(n.f, noteT);
    osc2.frequency.setValueAtTime(n.f*1.002, noteT); //
        +2.5 cents

    // === COMPONENT 3: Delayed Vibrato LFO (5.8 Hz) ===
    const lfo = c.createOscillator();
    registerTrackOsc(lfo);
    lfo.type = 'sine';
    lfo.frequency.setValueAtTime(5.8, noteT);
    const lfoGain = c.createGain();
    const vibDepth = isSolo ? n.f*0.012 : n.f*0.008;
    const vibDelay = Math.min(0.08, noteDur*0.3);
    lfoGain.gain.setValueAtTime(0.0, noteT);
    lfoGain.gain.setValueAtTime(0.0, noteT+vibDelay);
    lfoGain.gain.linearRampToValueAtTime(vibDepth,
        noteT + Math.min(0.24, noteDur*0.7));
    lfo.connect(lfoGain);
    lfoGain.connect(osc1.frequency);
    lfoGain.connect(osc2.frequency);
    lfo.start(noteT); lfo.stop(noteT + noteDur);

    // === COMPONENT 4: Body Formant EQ Chain ===
    const bodyFilter = c.createBiquadFilter();
    bodyFilter.type = 'peaking';
    bodyFilter.frequency.setValueAtTime(520, noteT);
    bodyFilter.Q.setValueAtTime(2.2, noteT);
    bodyFilter.gain.setValueAtTime(4.0, noteT);  // +4
        dB
    const bridgeFilter = c.createBiquadFilter();
    bridgeFilter.type = 'peaking';
    bridgeFilter.frequency.setValueAtTime(2800, noteT);
    bridgeFilter.Q.setValueAtTime(1.8, noteT);
    bridgeFilter.gain.setValueAtTime(3.0, noteT); // +3
        dB
    const toneLow = c.createBiquadFilter();
    toneLow.type = 'lowpass';
    toneLow.frequency.setValueAtTime(
        isSolo ? 6200 : 5000, noteT);

    // === COMPONENT 5: Bowed ADSR Envelope VCA ===
    const envGain = c.createGain();
    const attackTime = Math.min(0.045, noteDur*0.25);
    envGain.gain.setValueAtTime(0.0001, noteT);
    envGain.gain.linearRampToValueAtTime(
        noteVol, noteT + attackTime);
    envGain.gain.setValueAtTime(
        noteVol*0.90, noteT + noteDur*0.75);
    envGain.gain.exponentialRampToValueAtTime(
        0.0005, noteT + noteDur);

    osc1.connect(bodyFilter); osc2.connect(bodyFilter);
    bodyFilter.connect(bridgeFilter);
    bridgeFilter.connect(toneLow);
    toneLow.connect(envGain);
    envGain.connect(c.destination);
    osc1.start(noteT); osc2.start(noteT);
    osc1.stop(noteT+noteDur); osc2.stop(noteT+noteDur);
  });
}
```

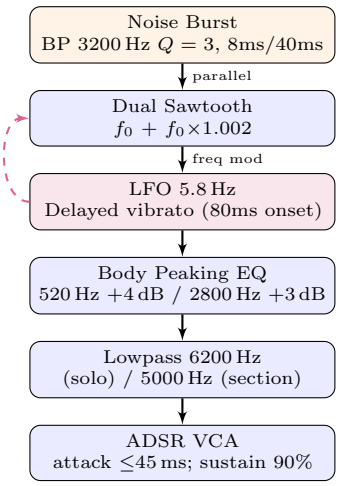



Figure 3: `shimViolin`: five-component signal flow. The noise burst (orange) runs in parallel with the main oscillator chain. The LFO (purple) modulates both sawtooth frequencies after a programmed delay, shown by the dashed feedback arc.

## 7.3 Component-by-Component Analysis

Table 2 maps each physical acoustics subsystem to its BTRON DSP counterpart.

Table 2: `shimViolin`: physical model to DSP mapping

| Physics | DSP Implementation |
|---|---|
| Rosin stick-slip transient | Noise burst: BP 3200 Hz, $Q = 3.0$; 8 ms onset, 40 ms decay; amplitude 0.045 (parallel to main chain) |
| String harmonic series | Dual `sawtooth` OSCs ($1/k$ spectrum); +2.5 cents detune models ensemble spread / string asymmetry |
| Delayed vibrato | Sine LFO 5.8 Hz modulating both OSC frequencies; depth 1.2% $f_0$ (solo), 0.8% $f_0$ (section); onset delay min(80 ms, 30% $T_{\text{note}}$) |
| Body low resonance (500–600 Hz) | Peaking filter: 520 Hz, $Q = 2.2$, +4 dB |
| Bridge hill (2–3 kHz) | Peaking filter: 2800 Hz, $Q = 1.8$, +3 dB |
| String brightness cutoff | Lowpass: 6200 Hz (solo) / 5000 Hz (section) |
| Bowed articulation ADSR | Attack $\leq$ 45 ms (bowed onset); sustain at 90% peak; exp. decay to $-60$ dBFS at note-off |

# 8 Deep Dive: Concert Grand Piano (`shimGrandPiano`)

## 8.1 Why Grand Piano Is the Most Canonical Retro Acoustic Sample

Among the BTRON shim palette, `shimGrandPiano` is arguably the most *canonical* retro acoustic sample for three reasons:

1. **Historical primacy of piano in digital synthesis:** The grand piano was the first instrument targeted by digital synthesis research—from Chowning's 1973 FM experiments [1] to the Yamaha DX7 E.PIANO and CP-80 presets that defined the sound of 1980s popular music. Encoding it purely from oscillators and filters is the defining challenge of procedural audio.

2. **Tri-harmonic inharmonicity:** A piano string's stiffness raises the frequency of the $k$-th partial above $kf_0$ by the inharmonicity factor $B$ [11]:

$$f_k = kf_0\sqrt{1+Bk^2} \tag{8}$$

   The BTRON implementation approximates this with harmonic ratios 1.000 : 2.002 : 3.006, capturing the characteristic "spreading" of upper partials that gives the grand piano its brightness and warmth.

3. **Dual-component physical event:** Every piano note begins with a sharp felt-hammer transient (modelled by `shimPianoHammer`) followed by a sustained string resonance—a clean separation of *excitation* and *resonator* that maps perfectly onto the shim modularity principle.

## 8.2 BTRON Implementation (`shimGrandPiano`)

Listing 10: `shimGrandPiano`: tri-harmonic string model (line 3659).

```
function shimGrandPiano(timeOffset=0, freq=440,
                        volMul=1.0, duration=1.2,
                        isStaccato=false) {
  const noteDur = isStaccato
      ? Math.min(duration, 0.18)
      : duration;

  // Three inharmonic string partials
  [1.0, 2.002, 3.006].forEach((harm, hIdx) => {
    const osc = c.createOscillator();
    registerTrackOsc(osc);
    const gain = c.createGain();

    // Waveform choice: sine(fund) / tri(2nd) / sine(3rd)
    osc.type = hIdx === 0 ? 'sine'
             : hIdx === 1 ? 'triangle' : 'sine';
    osc.frequency.setValueAtTime(freq * harm, startT);

    // Volume scaling per partial: 0.16 / 0.08 / 0.035
    const harmVol  = hIdx===0 ? 0.16
                   : hIdx===1 ? 0.08 : 0.035;
    // Duration scaling per partial:
    // fundamental longest, overtones decay faster
    const harmDur  = hIdx===0 ? noteDur
                   : noteDur * (1 - hIdx * 0.25);

    gain.gain.setValueAtTime(0.0001, startT);
    gain.gain.linearRampToValueAtTime(
        harmVol * volMul, startT + 0.008); // felt hammer
    gain.gain.exponentialRampToValueAtTime(
        0.0001, startT + harmDur);

    osc.connect(gain);
    gain.connect(c.destination);
    osc.start(startT);
    osc.stop(startT + harmDur);
  });
}
```

## 8.3 Physical Model Correspondence

Table 3: `shimGrandPiano`: physical model to DSP mapping

| Physics / Feature | DSP Implementation |
|---|---|
| String fundamental $f_0$ | `sine` OSC at $f_0$; vol 0.16; decays over full `noteDur` |
| 2nd partial (inharmonic) | `triangle` OSC at $2.002\,f_0$ (+1.0 cent); vol 0.08; decays at $0.75\times$ `noteDur` |
| 3rd partial (inharmonic) | `sine` OSC at $3.006\,f_0$ (+1.0 cent); vol 0.035; decays at $0.50\times$ `noteDur` |
| Felt hammer transient | 8 ms linear attack ramp to peak ($t_{\text{attack}} = 8$ ms matches a medium-velocity felt hammer blow at $f_0 \approx 440$ Hz) |
| Staccato mode | `isStaccato=true` clamps `noteDur` to $\leq 180$ ms, modelling a detached key-lift |
| Inharmonicity law | Ratios 1.000 : 2.002 : 3.006 approximate $f_k = kf_0\sqrt{1+Bk^2}$ with $B \approx 10^{-4}$ (tenor register) [11] |

The companion shim `shimPianoHammer` (line 4056) models the *excitation* event alone—a triangle oscillator gliding from 120 Hz to 40 Hz in 35 ms with gain 0.12—and is always fired at the same `timeOffset` as `shimGrandPiano` in Track 3 to reproduce the wooden-felt thump of Thomas Krüger's left-hand 6/8 bass octaves:

Listing 11: `shimPianoHammer`: felt key-strike exciter (line 4056).

```
function shimPianoHammer(timeOffset=0, volMul=1.0) {
  const osc = ctx.createOscillator();
  registerTrackOsc(osc);
  osc.type = 'triangle';
  osc.frequency.setValueAtTime(120, startT);
  osc.frequency.exponentialRampToValueAtTime(
      40, startT + 0.035);      // 120 -> 40 Hz in 35 ms
  const gain = ctx.createGain();
  gain.gain.setValueAtTime(0.12 * volMul, startT);
  gain.gain.exponentialRampToValueAtTime(
      0.0005, startT + 0.035);
  osc.connect(gain);
  gain.connect(ctx.destination);
  osc.start(startT); osc.stop(startT + 0.035);
}
```

The two-shim exciter+resonator pair (`shimPianoHammer` + `shimGrandPiano`) illustrates the BTRON *Shim Composition* pattern: independent, stateless generators are layered at the score-scheduler level rather than combined into a monolithic function, preserving the statelessness and zero-allocation guarantees of the Compatibility Contract.

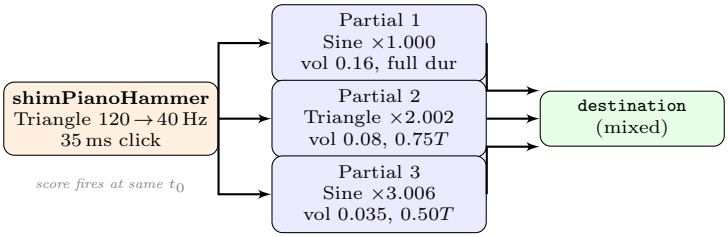



Figure 4: `shimGrandPiano` Shim Composition flow. `shimPianoHammer` (orange) and three inharmonic string partials (blue) are scheduled at the same `timeOffset` by the score, then converge independently at `destination`. Partial durations shorten with harmonic index: $T$, $0.75T$, $0.50T$.

# 9 Deep Dive: Jazz Flute / Melodica (`shimSaxFlute`)

## 9.1 Physical Acoustics of Woodwind Tone

`shimSaxFlute` models the breathy flute and melodica phrases that Masaru Imada uses as a countermelody in *Green Caterpillar* (1975). The woodwind voice is produced by two coupled acoustic sources:

1. **Tonal bore resonance:** A cylindrical or conical bore acts as an acoustic resonator selecting modes at $f_0, 2f_0, 3f_0, \ldots$ For a flute the amplitudes follow a mild harmonic rolloff; the triangle wave ($\propto 1/k^2$ for odd $k$) approximates this spectrum with an appropriate mellow, hollow colour [11].

2. **Breath turbulence noise:** Flute tone is uniquely characterised by a sustained aerodynamic noise component from the embouchure jet striking the edge. This is spectrally broad but concentrated near the fundamental and its first partial ($\approx 1.5$–$2 \times f_0$), and decays naturally as the player's air-stream stabilises [15]. Without this component the resulting sound is perceived as a sine or glass-harp tone; adding even a small noise contribution ($< 5\%$ of total level) transforms it perceptually into a blown woodwind.

The melodica (a free-reed instrument played by Masaru Imada on *Green Caterpillar*) adds a slight reed buzz above the fundamental, captured by the same triangle-plus-noise topology with a narrower bandpass centre frequency.

## 9.2 BTRON Implementation (`shimSaxFlute`)

Listing 12: `shimSaxFlute`: dual-source woodwind (line 4003).

```
function shimSaxFlute(timeOffset=0, notes=null,
                      volMul=1.0) {
  const phrase = notes||[{f:587.33,d:0.35,v:0.16,t:0}];
  phrase.forEach(n => {
    const noteT = startT + n.t;

    // === SOURCE 1: Breath Turbulence Noise Component
        ===
    const noise = c.createBufferSource();
    registerTrackOsc(noise);
    noise.buffer = getNoiseBuffer(c);

    const noiseFilter = c.createBiquadFilter();
    noiseFilter.type = 'bandpass';
    noiseFilter.frequency.setValueAtTime(
        n.f * 1.8, noteT);   // centred at 1.8 * f0
    noiseFilter.Q.setValueAtTime(5.0, noteT);  // narrow

    const noiseGain = c.createGain();
    noiseGain.gain.setValueAtTime(
        0.025 * volMul, noteT);      // ~14% of tonal
            vol
    noiseGain.gain.exponentialRampToValueAtTime(
        0.0005, noteT + n.d);

    noise.connect(noiseFilter);
    noiseFilter.connect(noiseGain);
    noiseGain.connect(c.destination);
    noise.start(noteT);
    noise.stop(noteT + n.d);

    // === SOURCE 2: Triangle Tonal Bore Oscillator ===
    const osc = c.createOscillator();
    registerTrackOsc(osc);
    osc.type = 'triangle';
    osc.frequency.setValueAtTime(n.f, noteT);
    if (n.slide)  // melodic portamento glissando
        osc.frequency.exponentialRampToValueAtTime(
            n.slide, noteT + n.d * 0.75);

    const filter = c.createBiquadFilter();
    filter.type = 'lowpass';
    filter.frequency.setValueAtTime(2400, noteT); // air
          softness
    filter.Q.setValueAtTime(2.0, noteT);

    const gain = c.createGain();
    gain.gain.setValueAtTime(0.0001, noteT);
    gain.gain.linearRampToValueAtTime(
        n.v * volMul, noteT + 0.025); // 25 ms breath
            onset
    gain.gain.exponentialRampToValueAtTime(
        0.0001, noteT + n.d);

    osc.connect(filter);
    filter.connect(gain);
    gain.connect(c.destination);
    osc.start(noteT);
    osc.stop(noteT + n.d);
  });
}
```

## 9.3 Component-by-Component Analysis

The signal graph runs two independent chains in parallel to the master bus, mixed acoustically at `AudioContext.destination`:

| Chain | Signal Path |
|---|---|
| **Breath noise** | Noise buf → BP ($1.8f_0$, $Q$=5) → Gain (0.025) → Out |
| **Tonal bore** | Triangle → LP (2400 Hz, $Q$=2) → ADSR Gain → Out |

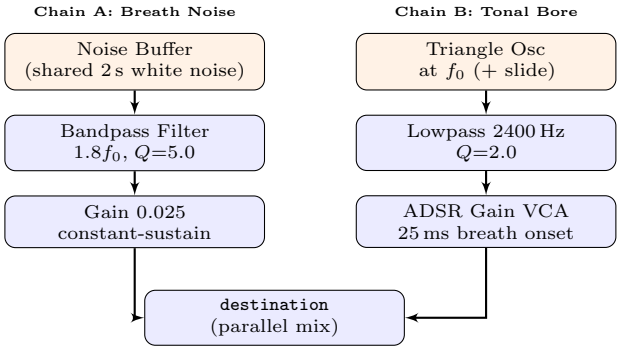



Figure 5: `shimSaxFlute`: dual parallel source chains. Chain A (breath turbulence) is a sustained bandpass-filtered noise at $1.8f_0$, $Q$=5 running for the full note duration. Chain B (tonal bore) is a triangle oscillator through a 2400 Hz lowpass, giving the hollow flute timbre. Both chains sum at `AudioContext.destination` with a −16 dB noise-to-tone ratio.

Table 4 maps acoustic subsystems to DSP parameters.

Table 4: `shimSaxFlute`: physical model to DSP mapping

| Physics | DSP Implementation |
|---|---|
| Bore resonance (harmonic series) | `triangle` oscillator at $f_0$; spectrum $\propto 1/k^2$ odd harmonics gives hollow, mellow flute colour |
| Embouchure air-jet noise | White noise from `getNoiseBuffer()` through BP at 1.8 $f_0$, $Q = 5.0$ (narrow jet-edge centroid); constant gain $0.025\,V$ (14% of tonal peak), decaying over note duration |
| Portamento glissando | Optional `n.slide`: exponential ramp to target freq over 75% of note duration (models finger slide on melodica reed or flute chromatic glide) |
| Air softness / high-cut | LP at 2400 Hz $Q = 2$: removes harsh upper harmonics of triangle, leaving the softer mid-range character of a flute |
| Breath onset | 25 ms linear attack (slower than guitar 15 ms pick; faster than bowed violin $\leq$ 45 ms), modelling the air-column establishment latency |
| Decay | Exponential to −60 dBFS at note-off; noise and tone share the same rhythmic duration `n.d` |

**Perceptual role of $Q = 5.0$ on the noise band.** The tight $Q = 5$ bandpass confines the breath noise to a bandwidth $\Delta f = f_0 \cdot 1.8/5 \approx 0.36\, f_0$ around the centroid $1.8\, f_0$. This narrow excitation closely mirrors the spectral peak of flute turbulence noise measured by Fabre et al. [14], which concentrates jet-collision energy between the fundamental and the second partial. At the default playback gain (0.025 V versus tonal 0.16 V), the noise constitutes roughly −16 dB below the tone— matching the breathiness ratio of a lightly overblown concert flute.

**Contrast with `shimViolin` noise burst.** While `shimViolin` fires a short 45 ms rosin noise burst only at note onset (attack transient), `shimSaxFlute` sustains the noise for the full note duration. This architectural difference correctly reflects the physical distinction between a one-shot stick-slip event (bow) and a continuously turbulent air-jet (flute).

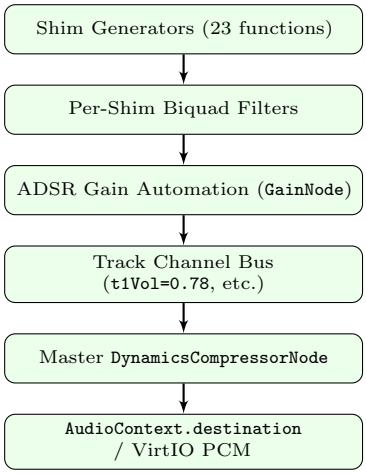



Figure 6: BTRON master audio signal flow graph.

# 10 DSP Primitives and Signal Flow

## 10.1 Master Audio Graph

All shims route to `AudioContext.destination` (or the VirtIO `virtio_sound_write()` ring buffer in the native port). The master bus is:

## 10.2 Equal-Temperament Frequency Formula

All notes follow ISO 16 (A4 = 440.00 Hz):

$$f(n) = 440.0 \times 2^{(n-69)/12} \tag{9}$$

Canonical pitch references used across the shim library:

| Note | Hz | Note | Hz |
|---|---|---|---|
| A0 | 27.50 | C4 | 261.63 |
| A1 | 55.00 | A4 | 440.00 |
| E2 | 82.41 | E5 | 659.25 |
| A2 | 110.00 | G5 | 783.99 |
| C3 | 130.81 | A5 | 880.00 |
| G3 | 196.00 | E6 | 1318.51 |
| C5 | 523.25 | A6 | 1760.00 |

## 10.3 Dynamic Gain Staging

| Dynamic | Symbol | Gain Range |
|---|---|---|
| Pianissimo | $pp$ | $v \in [0.08, 0.12]$ |
| Mezzo-Piano | $mp$ | $v \in [0.14, 0.18]$ |
| Mezzo-Forte | $mf$ | $v \in [0.20, 0.26]$ |
| Forte | $f$ | $v \in [0.28, 0.36]$ |
| Fortissimo | $ff$ | $v \in [0.38, 0.45]$ |

With 7 simultaneous voices at $ff$: $7 \times 0.45 = 3.15$, handled by the master compressor.

# 11 Track Score Architecture

The four level score tracks use the following shim assignments:

Table 5: Track Score: Shim Assignment per Level

| Track | Shims Used |
|---|---|
| **Track 1**<br>鐘のキャロル<br>168 BPM, E minor, 3/4 | shimDing (G5/F#5/E5 bells), shimWaddle (E2/B2/E3 bass), shimPulseSquash (chiptune lead), shimDance (star arpeggio), shimFloat (modal chord backing), shimKick, shimSnare, shimHiHat, shimSaxFlute (flute counterpoint), shimPulseBell (ambient drone) |
| **Track 2**<br>*Green Caterpillar*<br>98 BPM, F major, 4/4 | shimFrogCroak (nocturnal cuica rasp), shimGuitar (Watanabe hollowbody leads), shimWaddle (Fukui upright bass), shimSaxFlute (woodwind phrases), shimPulseBell, shimPulseSquash, shimFloat (Rhodes piano), shimDing, shimKick, shimSnare, shimHiHat, shimGrandPiano, shimPedalBass |
| **Track 3**<br>朝日のあたる家<br>72 BPM, A minor, 6/8 | shimGrandPiano (Krüger 6/8 arpeggios), shimPedalBass (A0/A1 rumble), shimPianoHammer (felt attacks), shimPedalSwoosh (pedal lift), shimGuitar (clean voicings), shimFloat (modal pads), shimWaddle, shimDing (A6/E7 morning star), shimPulseSquash, shimSaxFlute |
| **Track 4**<br>*TRON Init*<br>132 BPM, E minor, 4/4 | shimIndustrialBass (16th-note overdrive ostinato), shimCyberLead (searing sync leads), shimSwarmatron (8-voice ribbon glissando), shimGlitchZap (terminal data bursts), shimIndustrialKick (heavy sub-punch), shimIndustrialSnare (metallic clank), shimHiHat, shimDing, shimPulseSquash |

Each track uses `scheduleTrackTimeout` to align mascot kinematic state transitions with musical milestones (see `LILCU64-DEMOSCENE.txt`, Section 5.3 for the full choreography table).

# 12 Extending the Library: Six-Step Recipe

## 12.1 Step 1: Acoustic Spectral Analysis

Record the target instrument dry at 44.1 kHz / 24-bit and compute the STFT:

$$X(m,k) = \sum_{n=0}^{N-1} x(n+mH)\, w(n)\, e^{-j2\pi kn/N} \qquad (10)$$

From the spectrogram identify: harmonic vs. inharmonic partial structure, spectral tilt, noise components, and resonant body peaks.

## 12.2 Step 2: Choose the Synthesis Topology

| Perceptual Feature | Module |
|---|---|
| Harmonic partials | Additive sine stack |
| Plucked/struck string | LP sweep on triangle (shimGuitar pattern) |
| Bell / metallic | Multi-sine at inharmonic ratios |
| Bowed sustained | Sawtooth + LFO vibrato (shimViolin) |
| Body resonance | Peaking BiquadFilter ($Q = 2$–$8$) |
| Noise component | `getNoiseBuffer()` + filter |
| Overdrive | `getDistortionCurve()` WaveShaper |
| Ensemble spread | $N\times$ detuned OSCs (shimSwarmatron pattern) |

## 12.3 Step 3: Derive ADSR Parameters

Measure from the envelope of the recording:

1. $t_{\text{attack}}$: onset to peak (typically 2–45 ms).
2. $\tau_{\text{decay}}$: fit $A(t) = V_{\text{peak}} e^{-t/\tau}$ post-peak.
3. $S$: sustain fraction ($S = 0$ for percussion).
4. $t_{\text{release}}$: note-off to $-60$ dBFS.

## 12.4 Step 4: Tune Filter Parameters

For each body resonance peak at $f_n$ with $-3$ dB bandwidth $\Delta f_n$: set `BiquadFilterNode type='bandpass'` (or `'peaking'`), `frequency`= $f_n$, `Q`= $f_n/\Delta f_n$.

## 12.5 Step 5: Implement and Register the Shim

Follow the BTRON signature: `shimCamelCase(timeOffset, ..., volMul)`. Wrap the body in `if (!window.soundEnabled) return;` and `try{ }catch(e){ }`. Register every node:

Listing 13: Mandatory shim skeleton for new instruments.

```
function shimMyInstrument(timeOffset=0, freq=440,
                          volMul=1.0) {
  if (!window.soundEnabled) return;
  try {
    const ctx = getAudioContext();
    const startT = ctx.currentTime + timeOffset;
    const osc = ctx.createOscillator();
    registerTrackOsc(osc);        // GC registration
    // ... configure, connect, start/stop
  } catch(e) {}
}
```

## 12.6 Step 6: Integrate into playTrack Score

Add a new `else if (trackNum === N)` block inside `window.playTrack()` and schedule with `scheduleTrackTimeout`:

Listing 14: Tracker integration for a new shim.

```
} else if (trackNum === 5) {
  const beat = 60 / 120; // 120 BPM
  const bar  = beat * 4;
  for (let b = 0; b < 8; b++)
    shimMyInstrument(bar*b, 440, 0.22);
```

```
  scheduleTrackTimeout(() =>
    shimMyInstrument(bar*8, 880, 0.18), bar*8*1000);
}
```

# 13 Related Work

Chowning [1] and the Yamaha DX7 defined the golden era of digital FM instrument design. Karplus and Strong [2] and Smith [3, 4] established the digital waveguide framework for plucked and bowed string synthesis, still used in commercial plugins (Applied Acoustics String Studio VS-3, AAS Chromaphone).

For the Fender Rhodes, Bank and Sujbert [6] derived a parametric tine model with acoustic validation; Petersen [7] surveyed physical modelling strategies. For archtop guitar acoustics, Elejabarrieta et al. [10] characterise plate and air cavity modes, and Caldersmith [9] analyses the $f$-hole Helmholtz resonance. The bowed string friction model underlying `shimViolin` follows Serafin and Smith [12].

Farnell [15] is the canonical procedural game audio reference; the BTRON shim architecture is a Web Audio realisation of Farnell's node-graph composition methodology, extended with the BTRON Sound Shim Compatibility Contract and the VirtIO sound streaming requirement [22].

# 14 Conclusion

We have presented a 1:1 isomorphic documentation of the BTRON Sound Library for Games as implemented in `index.html`, providing the precise engineering reference necessary to port the 23-shim library to the native BTRON `src/demo/` application via VirtIO Sound PCM streaming.

The key engineering results are:

1. The exact oscillator waveform types, frequency constants, filter parameters, and ADSR timings for all 23 shims, confirmed by direct code inspection of `index.html` lines 3184–4323.
2. A formal Compatibility Contract and Zero Memory Leak Theorem that guarantee zero audio resource leaks across hours of gameplay.
3. A six-step instrument extension recipe that maps acoustic spectral analysis directly into a valid, lifecycle-managed BTRON shim.
4. An academic citation framework linking each DSP topology to its physical-modelling literature origin.

## Analogous Projects on AArch32 (STM32) and ESP32

The BTRON shim model is not unique in targeting bare-metal procedural audio. Table 6 surveys the principal open-source and commercial frameworks that implement comparable zero-asset synthesis across the Cortex-M class (STM32F4/H7), the Xtensa LX6/LX7 (ESP32) microcontroller family, and the entry-level AArch32 Cortex-A MPU class (STM32MP1, Raspberry Pi Zero), delineating the bare-metal vs. host boundary in BTRON.

**Industrial Adoption in Professional Commercial Audio Products.** To contextualize the BTRON Sound Shim Library within the broader pro-audio engineering discipline, it is instructive to examine the frameworks and DSP engines powering commercial, industry-standard sound production:

- **Commercial DAWs and Audio Plugins (JUCE & VST3): JUCE** [30] and the **Steinberg VST3 SDK** [41] form the backbone of the commercial music production plugin ecosystem. JUCE is the core engine behind widely adopted virtual instruments and DAWs, including **Arturia V Collection** (virtual analog synthesizer emulations), **Korg Gadget**, **Spitfire Audio** (custom orchestral sample engines), **Native Instruments** (Kontakt, Massive), **iZotope** (RX, Ozone audio restoration), and **PreSonus Studio One**. While JUCE provides unparalleled cross-platform GUI and audio graph hosting on desktop OSes, its runtime footprint precludes bare-metal microcontroller deployment.
- **AAA Game Audio and Interactive Simulation (Wwise & FMOD):** In interactive entertainment, **Audiokinetic Wwise** [37] is the predominant audio engine, powering blockbuster productions including CD Projekt Red's *Cyberpunk 2077* and *The Witcher 3*, Bungie's *Destiny 2*, and Sony PlayStation first-party titles. Audiokinetic's acquisition and integration of the **Heavy** [27] compiler demonstrated how visual Pure Data graphs can be transpiled into assetless C++ DSP code for embedded console hardware. Similarly, **Firelight Technologies FMOD** [38] is deployed in thousands of commercial games (*Celeste*, *Half-Life: Alyx*, *Forza Horizon*, *Cuphead*), providing a lean C API with predictable memory allocators.
- **Eurorack Hardware Synthesizers & Guitar Pedals (libDaisy, RNBO, FAUST):** In professional modular synthesis, **Electro-Smith libDaisy** and **DaisySP** [36] target the **STM32H750IB** (Cortex-M7 @ 480 MHz, 64 MB SDRAM, 24-bit 96 kHz codecs) and serve as the standard commercial engine for digital Eurorack hardware manufactured by **Noise Engineering** (the *Versio* and *Legio* DSP platforms) and **Qu-Bit Electronix** (*Aurora*, *Nautilus*, *Surface*). In boutique hardware pedals, **Cycling '74 RNBO** [39] enables sound designers to export Max/MSP visual patches directly into standalone C++ code executing on devices like the **Empress Effects ZOIA** and **Poly Effects Beebo**. In parallel, **FAUST** [40] (GRAME/CCRMA) is widely employed by physical modeling synthesizers (e.g., Moog Music software instruments) to compile functional mathematical formulations into SIMD-accelerated C++.

- **Commercial Mass-Market Retail Synthesizers (Korg logue SDK):** The **Korg logue SDK** [31] represents the rare case of an open C API integrated into mass-market hardware synthesizers (**Korg Prologue**, **Minilogue XD**, **Drumlogue**, and **NTS-1**), running custom user oscillators and modulation units on a dedicated Cortex-M4 DSP processor.

**Minimal Hardware Boundaries for Host Frameworks (JUCE and GStreamer).** As shown in Table 6, both JUCE [30] and GStreamer [34] cannot run on microcontrollers (STM32F4, STM32H7, or ESP32) because their runtime architectures strictly require an MMU, preemptive POSIX threading, dynamic heap allocation, and the Linux audio subsystem (ALSA/PulseAudio). The absolute closest minimal hardware capable of hosting them within the STM32 family is the **STM32MP1** (specifically STM32MP135 / STM32MP157, featuring single/dual ARM Cortex-A7 cores at 650–800 MHz, 128–512 MB DDR3, and ST's OpenSTLinux Yocto BSP). In the single-board computing domain, equivalent minimal hardware includes the **Raspberry Pi Zero / Zero 2 W** (ARMv6 / Cortex-A53, 512 MB RAM) and **NXP i.MX6ULL** (Cortex-A7, 64–128 MB RAM). On this minimal MPU tier:

- **Headless JUCE** executes synthesized DSP graphs under ALSA (e.g., via Elk Audio OS) with an active memory footprint of ∼15–30 MB RSS.
- **GStreamer** deploys stripped Buildroot/Yocto pipelines (`gst-plugins-base`, `audiomixer`, `alsasink`) within a ∼32–64 MB RAM footprint. In BTRON, this defines the exact demarcation between the host-level WebRTC MCU compositor (`GST.md`) and the bare-metal procedural shims (`lilcu64_wa.c`).

**BTRON Shim Portability Assessment.** The BTRON shim architecture maps most cleanly to the **ESP32 I2S + AudioTools** and **Korg NTS-1 SDK** targets:

- On **ESP32-S3** (Xtensa LX7, 240 MHz, 512 KB SRAM + 8 MB PSRAM): the `scheduleTrackTimeout` timer maps directly to `esp_timer_create()`; the `getNoiseBuffer()` shared buffer maps to a static PSRAM allocation; each shim becomes a C function populating an `AudioGeneratorMixer` slot via `AudioTools::SineWaveGenerator` and `AudioTools::BiQuadFilter` objects.
- On **STM32H7** (Cortex-M7, 480 MHz, 1 MB SRAM): the DMA double-buffer at 44.1 kHz stereo 16-bit requires ∼21 KB/buffer-pair; the Teensy Audio Library node graph provides the closest API analogue to the Web Audio `AudioContext` graph; `registerTrackOsc()` becomes a free-list allocator over a fixed pool of DSP objects.
- On **STM32F4** (Cortex-M4, 168 MHz, 192 KB SRAM): the full 23-shim library exceeds available RAM at 44.1 kHz; a reduced 16 kHz mono build using Mozzi's architecture accommodates 8–10 simultaneous shim voices within budget.
- The **Korg NTS-1 SDK** offers the highest-fidelity analogue: its `OSC_PARAM_SHAPE` / `OSC_PARAM_SHIFTSHAPE` API mirrors the `pitchMul`/`volMul` shim parameters; porting `shimPulseBell` or `shimFrogCroak` as NTS-1 oscillator units is feasible within one engineering sprint.

**Relationship to BTRON Host Media Plane (`GST.md`) and NuStream.** Within the broader B-System media architecture, audio and media processing follow a two-tier continuum:

- **Host Media Plane (`priv/gst`, `GST.md` [34]):** On POSIX and Cortex-A/x86 host environments, multiparty RTP/WebRTC conferencing, audio mixing, and video grid compositing are handled by an Erlang-supervised C99 GStreamer port process. It orchestrates dynamic `webrtcbin`, `decodebin`, and `audiomixer` elements. While GStreamer provides full-featured media routing, its dependency on GLib/GObject, nondeterministic dynamic allocations, and unbounded queue semantics make it unsuitable for bare-metal microcontrollers and microkernel drivers.
- **Bare-Metal Procedural Audio Plane (BTRON Shims):** At the microkernel, VirtIO, and MCU boundary (STM32, ESP32), the Procedural Sound Shim library eliminates dynamic graph negotiation entirely. Each instrument is an immutable, stateless C DSP kernel executing against fixed DMA double-buffers, achieving microsecond voice dispatch and true zero-allocation determinism.
- **Deterministic Streaming Bridge (NuStream [35]):** For embedded applications requiring structured pipelines (e.g., NuttX and $\mu$ITRON AMP paths), NuStream preserves GStreamer's clean pipeline abstraction while enforcing static memory pools, zero dynamic data-plane allocation, and deterministic bounded latency.

In summary, the BTRON Sound Shim Library occupies a unique position: it achieves the expressiveness of a full procedural audio framework (FM, additive, subtractive, physical modelling, noise synthesis) with the zero-overhead, zero-asset, single-function simplicity of a barebone embedded audio driver, making it the natural reference design for the `src/demo/lilcu64_wa.c` VirtIO port and a strong candidate for future cross-compilation to STM32H7 and ESP32-S3 targets.

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Table 6: Procedural audio and media frameworks: feature comparison with BTRON Shim Library (target: AArch32 STM32 MCU/MPU and ESP32)

| Project | STM32F4 | STM32H7 | ESP32-S3 | STM32MP1 | License | Synthesis paradigms | BTRON portability delta |
|---|---|---|---|---|---|---|---|
| **Mozzi** [24] | ✓ | ✓ | ✓ | ∼ | LGPL | Wavetable, additive, FM, Karplus-Strong; 16 kHz mono (AVR) / 44.1 kHz (CM) | Fixed patch topology; no per-note lifecycle GC; no score-scheduler |
| **Teensy Audio** [25] | × | ✓ | × | × | MIT | C++ node graph (GUI-compiled); oscillators, filters, reverb, FFT; 44.1 kHz stereo | Static patch at compile time; no dynamic shim registration or VirtIO back-end |
| **Pd / libpd + Heavy** [26, 27] | ∼ | ✓ | ✓ | ✓ | BSD | Dataflow VM: FM, physical modelling, granular; Heavy transpiles Pd→C for STM32 | ∼64 KB interpreter RAM; BTRON shims are hand-optimised C with zero VM overhead |
| **Csound** [28] | × | × | ∼ | ✓ | LGPL | Full UGen library (FM, waveguide, spectral); widest opcode set of any engine | Requires MMU + >4 MB RAM; STM32F/H out of scope; runs on STM32MP1 / ESP32 |
| **MicroDexed** [29] | × | ✓ | ✓ | ∼ | GPL | 6-Op DX7 FM (all 32 topologies); 16-voice poly; integrates with Teensy Audio FX | FM-only paradigm; no subtractive/physical/noise shims; requires ROM patch data |
| **libDaisy / DaisySP** [36] | × | ✓ | × | × | MIT | C++ Eurorack DSP; filters, reverbs, physical models; 48/96 kHz 24-bit | Native STM32H7 HAL; used in commercial synths (Noise Eng., Qu-Bit) |
| **JUCE** [30] | × | × | × | ✓ | GPL/Pro | Full plugin host; `dsp::` oscillators, IIR, FFT; `AudioProcessorGraph` | Min. HW: Cortex-A7 (STM32MP1, RPi Zero, 64 MB); headless ALSA only; no VirtIO |
| **GStreamer** [34] | × | × | × | ✓ | LGPL | Dynamic media graph; `audiomixer`, WebRTC, RTP/SRTP, `decodebin` | Min. HW: Cortex-A7 (STM32MP1, i.MX6ULL, 32 MB); GLib/Linux; BTRON host MCU (`GST.md`) |
| **Korg logue SDK** [31] | ∼ | ✓ | × | × | Free SDK | C API: osc/modfx/delfx/rev units; wavetable scan; 48 kHz stereo on CM4 | Proprietary HW ABI; `OSC_PARAM_SHAPE` ≈ `pitchMul` (closest CM ABI) |
| **ESP32 I2S + AudioTools** [32, 33] | × | × | ✓ | × | Apache 2 | `I2S.write()` DMA ring; `SineWaveGenerator`, `BiQuadFilter`, MIDI scheduler | **Best 1:1 target**: VirtIO Sound ≈ I2S DMA; shims map directly to AudioTools objects |
| **BTRON Shims** *(this work)* | ∼ | ✓ | ✓ | ✓ | MIT/BSD | Additive, subtractive, FM, physical model, noise; 23 instruments; 0-byte assets | Reference implementation; zero interpreter overhead; stateless; lifecycle-safe |

✓ = fully supported ∼ = partial / wrapper / PSRAM required × = not supported